

Digital Logic Circuits (Spring 2026)

HW #5 (Due: May 28, 2026)

4. For the ripple counter in Figure 9–66, show the complete timing diagram for sixteen clock pulses. Show the clock, Q_0 , Q_1 , and Q_2 waveforms.

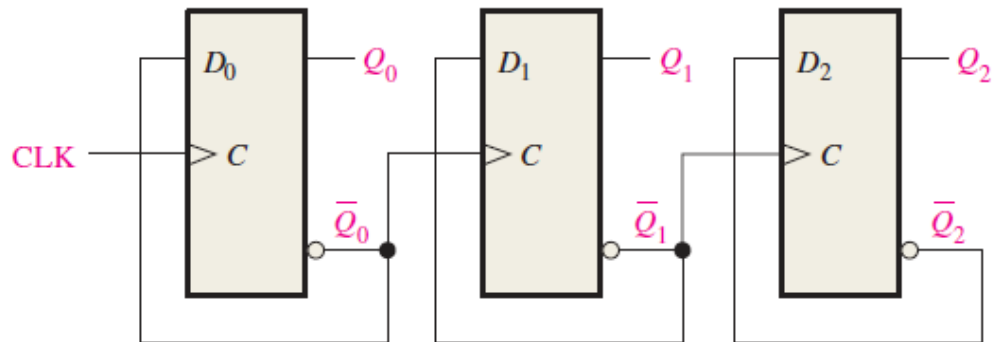


FIGURE 9–66

5. In the counter of Problem 4, assume that each flip-flop has a propagation delay from the triggering edge of the clock to a change in the Q output of 8 ns. Determine the worst-case (longest) delay time from a clock pulse to the arrival of the counter in a given state. Specify the state or states for which this worst-case delay occurs.
8. Show the complete timing diagram for the 5-stage synchronous binary counter in Figure 9–67. Verify that the waveforms of the Q outputs represent the proper binary number after each clock pulse.

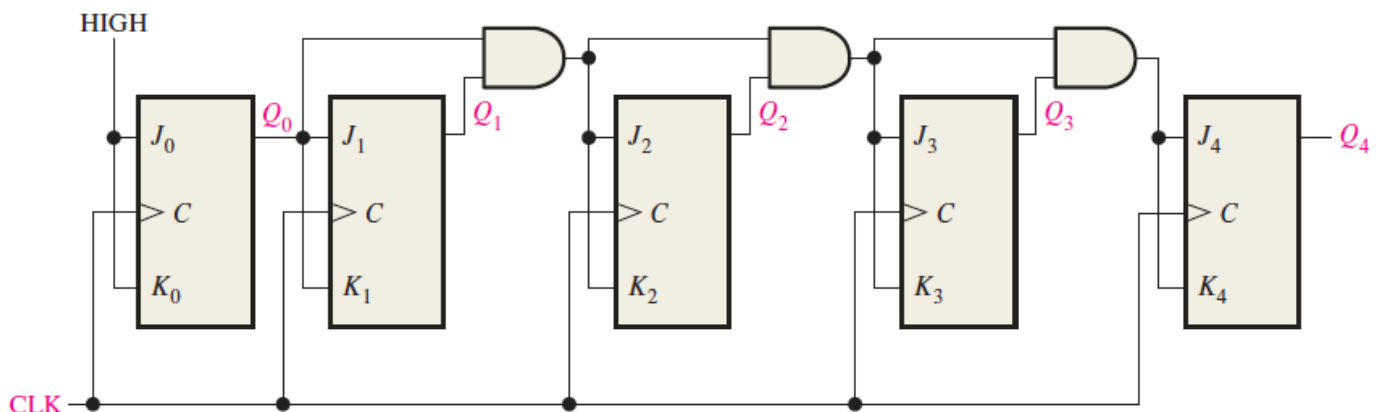


FIGURE 9–67

18. Determine the sequence of the counter in Figure 9–73.

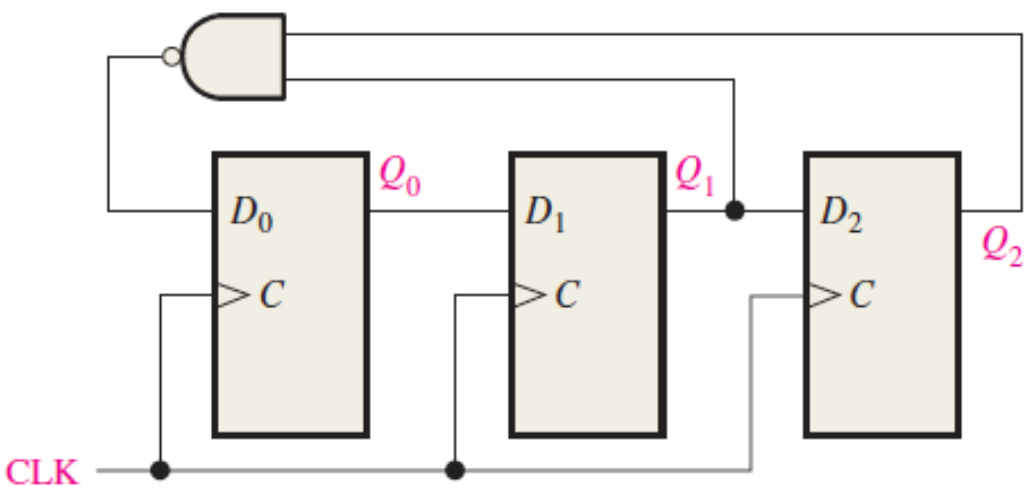


FIGURE 9–73

21. Design a counter to produce the following binary sequence. Use J-K flip-flops.
1, 4, 3, 5, 7, 6, 2, 1, . . .

9. Determine the output of the array in Figure 10–67. The Xs represent connected links.

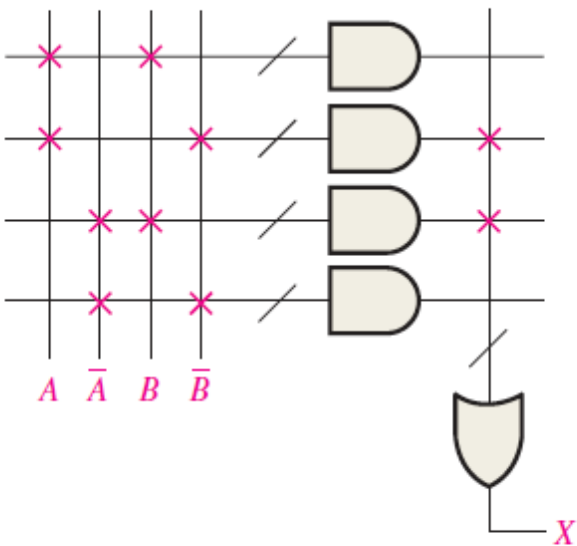


FIGURE 10–67

10. Modify the array in Figure 10–67 to produce an output $X = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + ABC + A\overline{B}C$

16. Generally, what elements make up a configurable logic block (CLB) in an FPGA? What elements make up a logic module?

12. Determine the truth table for the ROM in Figure 11–80.

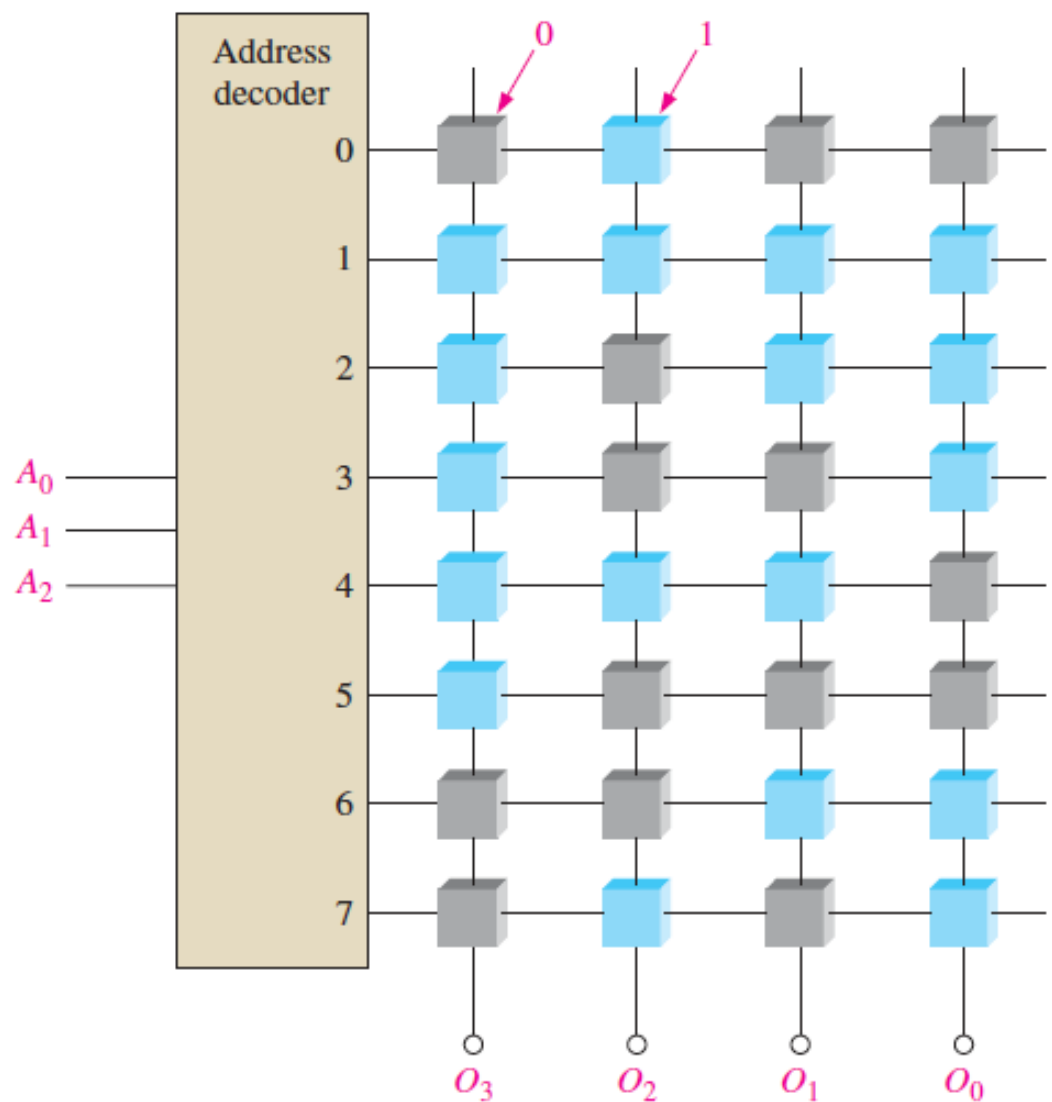


FIGURE 11–80

17. Use $16k \times 4$ DRAMs to build a $64k \times 8$ DRAM. Show the logic diagram.

4. For the ripple counter in Figure 9-66, show the complete timing diagram for sixteen clock pulses. Show the clock, Q_0 , Q_1 , and Q_2 waveforms.

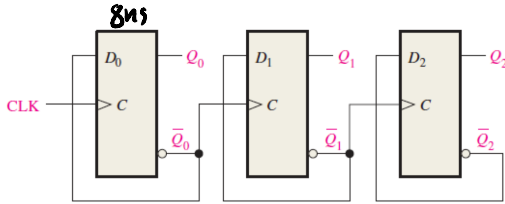
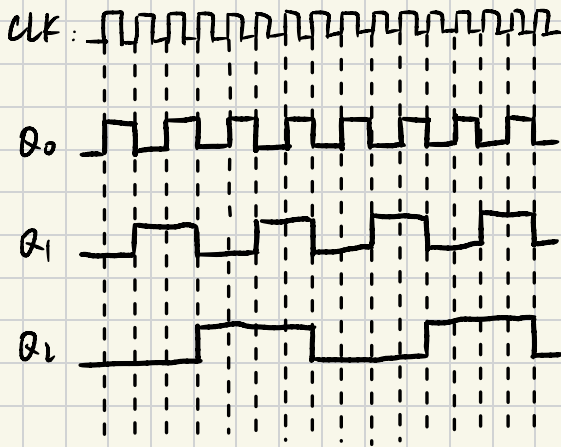


FIGURE 9-66



5. In the counter of Problem 4, assume that each flip-flop has a propagation delay from the triggering edge of the clock to a change in the Q output of 8 ns. Determine the worst-case (longest) delay time from a clock pulse to the arrival of the counter in a given state. Specify the state or states for which this worst-case delay occurs.

$$t_{prop\ max} = (8 + 8 + 8) \text{ ns} = 24 \text{ ns}$$

Worst: $Q_2 Q_1 Q_0$ 000 $\rightarrow$ 111 or 111 $\rightarrow$ 000

and those shifts requiring for 3-bit changes

8. Show the complete timing diagram for the 5-stage synchronous binary counter in Figure 9-67. Verify that the waveforms of the Q outputs represent the proper binary number after each clock pulse.

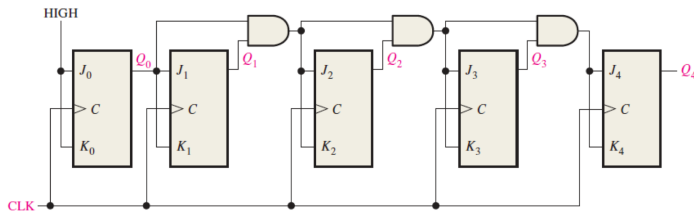
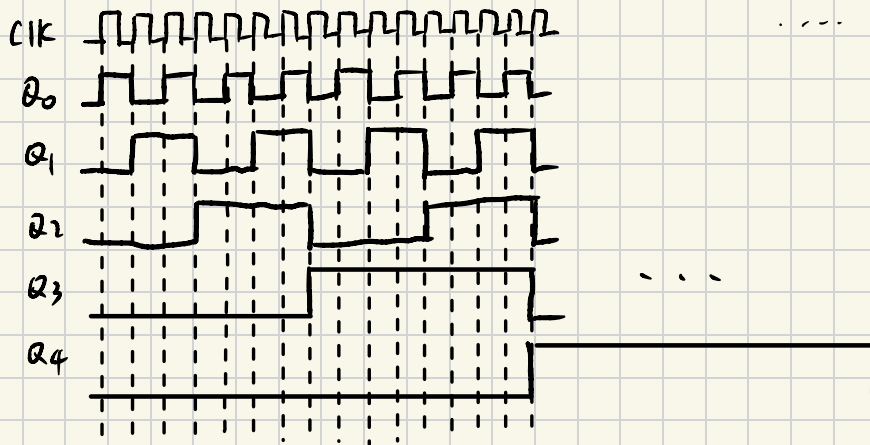


FIGURE 9-67



$Q_4 Q_3 Q_2 Q_1 Q_0$:

00000
00001
00010
00011
00100
00101
...

18. Determine the sequence of the counter in Figure 9-73.

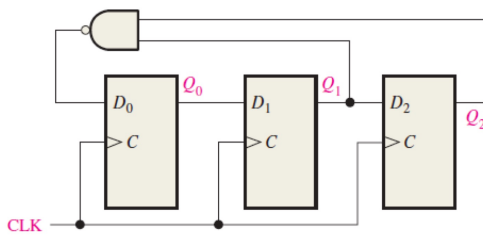


FIGURE 9-73

$Q_2 Q_1 Q_0$:

0 0 0
0 0 1
0 1 1
1 1 1
1 1 0
1 0 0
0 0 1
0 1 1
1 1 1
1 1 0
1 0 0
...

period sequence

21. Design a counter to produce the following binary sequence. Use J-K flip-flops.

1, 4, 3, 5, 7, 6, 2, 1, ...

$Q_2 Q_1 Q_0$	$Q_2 Q_1 Q_0 \rightarrow Q_2 Q_1 Q_0$	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$
1 001	0 0 1	1 X	0 X	X 1
4 100	1 0 0	X 1	1 X	1 X
3 011	0 1 1	1 X	X 1	X 0
5 101	1 0 1	X 0	1 X	X 0
7 111	1 1 1	X 0	X 0	X 1
6 110	1 1 0	X 1	X 0	0 X
2 010	0 1 0	0 X	X 1	1 X
1 001				

J_0 :	$Q_2 \backslash Q_1 Q_0$	00	01	11	10
	0	X	X	X	1
	1	1	X	X	0
		$= \bar{Q}_2 + \bar{Q}_1$			

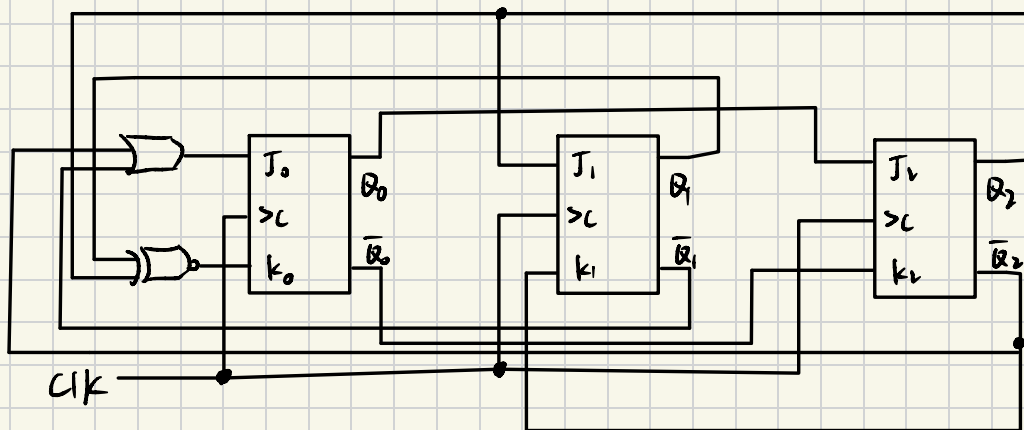
K_0 :	$Q_2 \backslash Q_1 Q_0$	00	01	11	10
	0	X	1	0	X
	1	X	0	1	X
		$= \bar{Q}_2 \bar{Q}_1 + Q_2 Q_1$			

J_1 :	$Q_2 \backslash Q_1 Q_0$	00	01	11	10
	0	X	0	X	0
	1	1	1	X	X
		$= Q_2$			

K_1 :	$Q_2 \backslash Q_1 Q_0$	00	01	11	10
	0	X	X	1	X
	1	X	X	0	0
		$= \bar{Q}_2$			

J_2 :	$Q_2 \backslash Q_1 Q_0$	00	01	11	10
	0	X	1	1	0
	1	X	X	X	X
		$= Q_0$			

K_2 :	$Q_2 \backslash Q_1 Q_0$	00	01	11	10
	0	X	X	X	X
	1	1	0	0	1
		$= \bar{Q}_0$			



9. Determine the output of the array in Figure 10-67. The Xs represent connected links.

$$9. \bar{A}\bar{B} + \bar{A}B$$

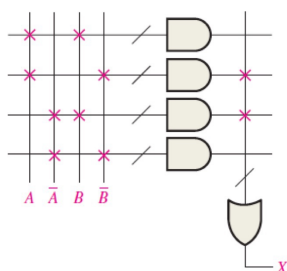
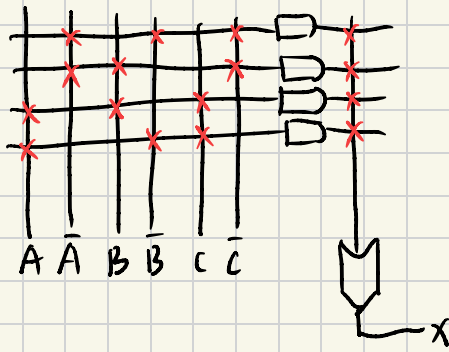


FIGURE 10-67

10. Modify the array in Figure 10-67 to produce an output $X = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + ABC + A\overline{B}C$

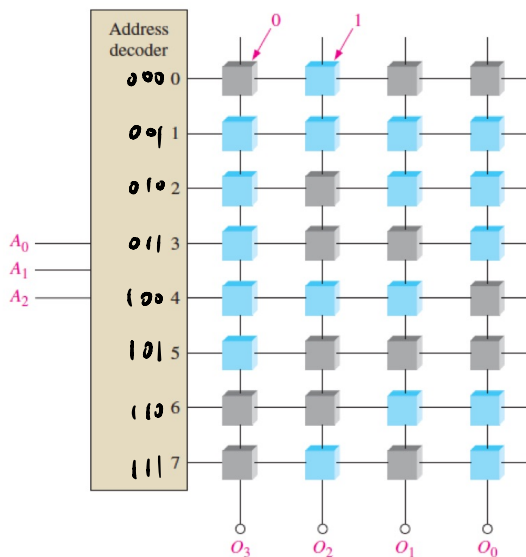


16. Generally, what elements make up a configurable logic block (CLB) in an FPGA? What elements make up a logic module?

In a FPGA, CLBs are made up of logic modules

Each logic module is made up of LUT, flip-flop and Multiplexers, or other associative logics.

12. Determine the truth table for the ROM in Figure 11-80.



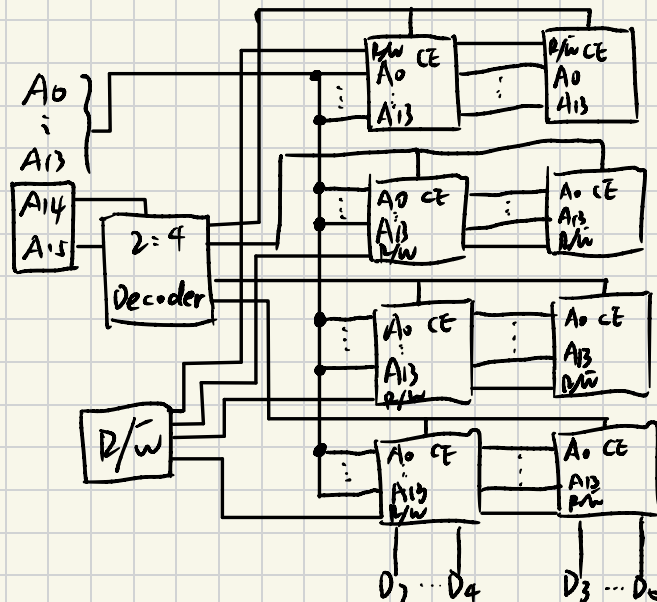
A_2	A_1	A_0	O_3	O_2	O_1	O_0
0	0	0	0	1	0	0
0	0	1	1	1	1	1
0	1	0	1	0	1	1
0	1	1	1	0	0	1
1	0	0	1	1	1	0
1	0	1	1	0	0	0
1	1	0	0	0	1	1
1	1	1	0	1	0	1

FIGURE 11-80

17. Use $16k \times 4$ DRAMs to build a $64k \times 8$ DRAM. Show the logic diagram.

$$64k / 16k = 4, \quad 8 / 4 = 2$$

4x2



Hint: $A_0-A_{13}/R/\overline{W}$ connect to 2 chip in parallel, not serial.

Just for easier recognition.